

G1 Probability — Answer Key

IGCSE Mathematics 0580 | Full Solutions

G1_Q1: Basic Probability - Problems

F Foundation — Answer: $3/8$

Total balls = $3 + 5 = 8$. $P(\text{red}) = \text{number of red} / \text{total} = 3/8$.

F Foundation — Answer: $1/6$

There is 1 favourable outcome (5) out of 6 equally likely outcomes. $P(5) = 1/6$.

F Foundation — Answer: 0.7

$P(\text{not rain}) = 1 - P(\text{rain}) = 1 - 0.3 = 0.7$.

F Foundation — Answer: $1/4$

There is 1 yellow section out of 4 equal sections. $P(\text{yellow}) = 1/4$.

C Core — Answer: $1/3$

Total pens = $7 + 5 + 3 = 15$. $P(\text{blue}) = 5/15 = 1/3$.

C Core — Answer: $1/3$

Numbers greater than 4 on a dice are 5 and 6: 2 favourable outcomes. Total outcomes = 6. $P(>4) = 2/6 = 1/3$.

C Core — Answer: $3/5$

Total marbles = $2 + 6 + 7 = 15$. Non-white marbles = $2 + 7 = 9$. $P(\text{not white}) = 9/15 = 3/5$.

C Core — Answer: $1/2$

There are 13 hearts and 13 diamonds, 26 favourable cards total. $P(\text{heart or diamond}) = 26/52 = 1/2$.

C Core — Answer: $2/11$

The word MATHEMATICS has 11 letters. M appears twice. $P(M) = 2/11$.

C Core — Answer: 0.3

$P(\text{yellow}) = 1 - P(\text{red}) - P(\text{blue}) = 1 - 0.2 - 0.5 = 0.3$.

E Unknown — Answer: $1/6$

Total outcomes = $6 \times 6 = 36$. Ways to sum to 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — 6 ways. $P(\text{sum}=7) = 6/36 = 1/6$.

E Unknown — Answer: 1/9

Total outcomes = 36. Pairs with product 12: (2,6), (3,4), (4,3), (6,2) — 4 ways. $P = 4/36 = 1/9$.

E Unknown — Answer: 2044/3375

Total marbles = 15. $P(\text{not red in one draw}) = (5+6)/15 = 11/15$. $P(\text{no red in three draws}) = (11/15)^3 = 1331/3375$. $P(\text{at least one red}) = 1 - 1331/3375 = (3375-1331)/3375 = 2044/3375$.

E Unknown — Answer: 1/4

Even numbers on 1-8 are {2,4,6,8}: 4 out of 8 = $1/2$. $P(\text{both even}) = (1/2) \times (1/2) = 1/4$.

X Extension — Answer: 5/54

Total outcomes = $6^3 = 216$. Choose 3 distinct numbers from $\{1, \dots, 6\} = C(6,3) = 20$. For each chosen set, there is exactly 1 strictly increasing order. $P = 20/216 = 5/54$.

X Extension — Answer: 35/66

Total balls = 12. Total ways to choose 2 = $C(12,2) = 66$. $P(\text{both white}) = C(5,2)/66 = 10/66$. $P(\text{both black}) = C(7,2)/66 = 21/66$. $P(\text{different colours}) = 1 - 10/66 - 21/66 = (66-31)/66 = 35/66$.

G1_Q2: Sample Space Diagrams - Problems

F Foundation — Answer: HH, HT, TH, TT

Each coin can be H or T. The sample space is {HH, HT, TH, TT}.

F Foundation — Answer: 1/2

Sample space: {HH, HT, TH, TT}. Outcomes with exactly one head: HT, TH = 2 outcomes. $P = 2/4 = 1/2$.

F Foundation — Answer: 1/6

Sample space: {(1,H),(1,T),(2,H),(2,T),(3,H),(3,T)} — 6 outcomes. Only (3,H) is favourable. $P = 1/6$.

F Foundation — Answer: 4

Total 5 appears at (1,4), (2,3), (3,2), (4,1) — 4 outcomes.

C Core — Answer: 1/6

Total outcomes = 36. Doubles: (1,1),(2,2),(3,3),(4,4),(5,5),(6,6) — 6 outcomes. $P(\text{double}) = 6/36 = 1/6$.

C Core — Answer: 5/12

Total outcomes = 36. Sum ≥ 8 : 15 outcomes. $P = 15/36 = 5/12$.

C Core — Answer: 1/4

Sample space has 12 outcomes. Favourable: (H,2),(H,4),(H,6) — 3 outcomes. $P = 3/12 = 1/4$.

C Core — Answer: 5/12

Total outcomes = 36. Prime sums: 2,3,5,7,11. Sum=2: 1 way. Sum=3: 2 ways. Sum=5: 4 ways. Sum=7: 6 ways. Sum=11: 2 ways. Total = 15. $P = 15/36 = 5/12$.

C Core — Answer: 5/12

$P(\text{first} > \text{second}) = P(\text{first} < \text{second})$. $P(\text{first} = \text{second}) = 6/36 = 1/6$. So $P(\text{first} > \text{second}) = (1 - 1/6)/2 = 5/12$.

C Core — Answer: 1/3

Total outcomes = 36. Multiples of 3 sum: 3(2 ways), 6(5 ways), 9(4 ways), 12(1 way). Total = 12. $P = 12/36 = 1/3$.

E Unknown — Answer: 1/6

Total outcomes = 36. Difference = 3: (1,4),(2,5),(3,6),(4,1),(5,2),(6,3) — 6 ways. $P = 6/36 = 1/6$.

E Unknown — Answer: 5/6

Total outcomes = 36. Sum > 9 means sum = 10(3 ways), 11(2 ways), 12(1 way): total 6. $P(\text{sum} \leq 9) = 1 - 6/36 = 30/36 = 5/6$.

E Unknown — Answer: 1/2

Given exactly one red: outcomes are RW and WR. Of these, 1 has first card red (RW). $P = 1/2$.

E Unknown — Answer: 5/18

Total outcomes = 36. Pairs with product ≥ 18 : (3,6),(4,5),(4,6),(5,4),(5,5),(5,6),(6,3),(6,4),(6,5),(6,6) — 10 outcomes. $P = 10/36 = 5/18$.

X Extension — Answer: 1/8

Total outcomes = $6^3 = 216$. Count ordered triples with sum 10: $a=1:4$, $a=2:5$, $a=3:6$, $a=4:5$, $a=5:4$, $a=6:3$. Total = 27. $P = 27/216 = 1/8$.

X Extension — Answer: 9/16

Total outcomes = $4 \times 4 = 16$. Prime sums: 2(1 way), 3(2 ways), 5(4 ways), 7(2 ways). Total = 9. $P = 9/16$.

G1_Q3: Tree Diagrams - Problems

F Foundation — Answer: 1/4

First toss: H(1/2) or T(1/2). Second toss: H(1/2) or T(1/2). $P(HH) = 1/2 \times 1/2 = 1/4$.

F Foundation — Answer: 1/16

$P(3) = 1/4$, $P(1) = 1/4$, independent. $P(3 \text{ then } 1) = 1/4 \times 1/4 = 1/16$.

F Foundation — Answer: 4/25

$P(\text{red}) = 2/5$ each time, independent. $P(RR) = 2/5 \times 2/5 = 4/25$.

F Foundation — Answer: 1/3

$P(RR) = 1/3 \times 1/3 = 1/9$. Similarly $P(BB) = 1/9$, $P(GG) = 1/9$. $P(\text{same}) = 1/9 + 1/9 + 1/9 = 3/9 = 1/3$.

C Core — Answer: 0.48

$P(\text{rain}) = 0.4$, $P(\text{dry}) = 0.6$. $P(RD) = 0.4 \times 0.6 = 0.24$, $P(DR) = 0.6 \times 0.4 = 0.24$. $P(\text{exactly one}) = 0.24 + 0.24 = 0.48$.

C Core — Answer: 3/10

$P(\text{first red}) = 3/5$. After removing 1 red: 2 red, 2 blue remain. $P(\text{second red} \mid \text{first red}) = 2/4 = 1/2$. $P(\text{both red}) = 3/5 \times 1/2 = 3/10$.

C Core — Answer: 3/5

$P(RB) = 3/5 \times 2/4 = 6/20 = 3/10$. $P(BR) = 2/5 \times 3/4 = 6/20 = 3/10$. $P(\text{different}) = 3/10 + 3/10 = 6/10 = 3/5$.

C Core — Answer: 0.512

$P(\text{germinate}) = 0.8$, independent. $P(\text{all three}) = 0.8 \times 0.8 \times 0.8 = 0.512$.

C Core — Answer: 0.42

$P(WL) = 0.3 \times 0.7 = 0.21$. $P(LW) = 0.7 \times 0.3 = 0.21$. $P(\text{exactly one}) = 0.21 + 0.21 = 0.42$.

C Core — Answer: 7/15

$P(RR) = 4/10 \times 3/9 = 12/90 = 2/15$. $P(BB) = 6/10 \times 5/9 = 30/90 = 1/3 = 5/15$. $P(\text{same}) = 2/15 + 5/15 = 7/15$.

E Unknown — Answer: 1/15

Total = 10. $P(\text{first green}) = 3/10$. After removing 1 green: 2 green left. $P(\text{second green} \mid \text{first green}) = 2/9$. $P(\text{both green}) = 3/10 \times 2/9 = 6/90 = 1/15$.

E Unknown — Answer: 31/45

$P(\text{same}) = P(RR) + P(GG) + P(YY)$. $P(RR) = 5/10 \times 4/9 = 20/90 = 2/9$. $P(GG) = 3/10 \times 2/9 = 6/90 = 1/15$. $P(YY) = 2/10 \times 1/9 = 2/90 = 1/45$. $P(\text{same}) = 20/90 + 6/90 + 2/90 = 28/90 = 14/45$. $P(\text{different}) = 1 - 14/45 = 31/45$.

E Unknown — Answer: 5/6

$P(\text{all black}) = 6/10 \times 5/9 \times 4/8 = 120/720 = 1/6$. $P(\text{at least one white}) = 1 - 1/6 = 5/6$.

E Unknown — Answer: 7/9

After removing 1 red: 2 red, 7 blue remain. $P(\text{second blue} \mid \text{first red}) = 7/9$.

X Extension — Answer: 1/4

Total = 10. Each ordering of R,W,G: $P = 2/10 \times 3/9 \times 5/8 = 30/720 = 1/24$. Number of orderings = $3! = 6$. Total = $6 \times 1/24 = 6/24 = 1/4$.

X Extension — Answer: 0.38

$P(M,P,\text{not } C) = 0.6 \times 0.5 \times 0.6 = 0.18$. $P(M,\text{not } P,C) = 0.6 \times 0.5 \times 0.4 = 0.12$. $P(\text{not } M,P,C) = 0.4 \times 0.5 \times 0.4 = 0.08$. Total = $0.18 + 0.12 + 0.08 = 0.38$.

G1_Q4: Conditional Probability - Problems

F Foundation — Answer: 2/7

After removing 1 red: 2 red, 5 blue remain. $P(\text{second red} \mid \text{first red}) = 2/7$.

F Foundation — Answer: 1/3

Even numbers: {2,4,6}. Of these, only 6 is favourable. $P(6 \mid \text{even}) = 1/3$.

F Foundation — Answer: 1/2

Red cards: 26 total (13 hearts + 13 diamonds). Hearts: 13. $P(\text{heart} \mid \text{red}) = 13/26 = 1/2$.

F Foundation — Answer: 5/9

Number of girls = 18. Girls who wear glasses = 10. $P(\text{glasses} \mid \text{girl}) = 10/18 = 5/9$.

C Core — Answer: 5/12

$P(\text{French}) = 60/100 = 3/5$. $P(\text{both}) = 25/100 = 1/4$. $P(\text{German} \mid \text{French}) = P(\text{both})/P(\text{French}) = (1/4)/(3/5) = 1/4 \times 5/3 = 5/12$. Or directly: $25/60 = 5/12$.

C Core — Answer: 4/9

After removing 1 blue: 4 red, 5 blue remain. $P(\text{second red} \mid \text{first blue}) = 4/9$.

C Core — Answer: 1/6

Ways to sum to 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) — 6 equally likely outcomes. Of these, 1 has first dice = 3: (3,4). $P = 1/6$.

C Core — Answer: 1/3

There are 12 face cards (3 per suit $\times$ 4 suits). Kings: 4. $P(\text{King} \mid \text{face card}) = 4/12 = 1/3$.

C Core — Answer: 5/9

Number who passed = 63. Girls who passed = 35. $P(\text{girl} \mid \text{passed}) = 35/63 = 5/9$.

C Core — Answer: 5/8

Non-green marbles = $5 + 3 = 8$. Red marbles among those = 5. $P(\text{red} \mid \text{not green}) = 5/8$.

E Unknown — Answer: 1/2

$P(RB) = 3/10 \times 7/9 = 21/90$. $P(BR) = 7/10 \times 3/9 = 21/90$. $P(\text{different}) = 42/90$. $P(\text{first red} \mid \text{different}) = (21/90)/(42/90) = 21/42 = 1/2$.

E Unknown — Answer: 95/242

$P(D)=0.02$, $P(\text{no } D)=0.98$. $P(T+|D)=0.95$, $P(T+|\text{no } D)=0.03$. $P(T+)=0.02 \times 0.95 + 0.98 \times 0.03 = 0.019 + 0.0294 = 0.0484$. $P(D|T+)=0.019/0.0484=190/484=95/242$.

E Unknown — Answer: 1/5

$P(\text{both red})=4/10 \times 3/9=12/90=2/15$. $P(\text{at least one red})=1-P(\text{both white})=1-(6/10 \times 5/9)=1-30/90=2/3$. $P(\text{both red} \mid \text{at least one red})=(2/15)/(2/3)=2/15 \times 3/2=3/15=1/5$.

E Unknown — Answer: 2/3

Prime numbers on a dice: $\{2,3,5\}$. Of these, odd numbers are $\{3,5\}$. $P(\text{odd} \mid \text{prime}) = 2/3$.

X Extension — Answer: 4

$P(RR) = 6/(6+n) \times 5/(5+n) = 1/3$. So $30/((6+n)(5+n)) = 1/3$. $90 = (6+n)(5+n) = n^2 + 11n + 30$. $n^2 + 11n - 60 = 0$. $(n+15)(n-4)=0$. $n=4$.

X Extension — Answer: 3/8

$P(A)=0.6$, $P(B)=0.4$. $P(\text{def}|A)=0.02$, $P(\text{def}|B)=0.05$. $P(\text{def})=0.6 \times 0.02 + 0.4 \times 0.05 = 0.012 + 0.02 = 0.032$. $P(A|\text{def})=0.012/0.032=12/32=3/8$.

G1_Q5: Venn Diagrams & Probability - Problems

F Foundation — Answer: 10

$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 5 + 8 - 3 = 10$.

F Foundation — Answer: 1/5

Football only = $12 - 4 = 8$. Tennis only = $8 - 4 = 4$. Both = 4. Neither = $20 - (8 + 4 + 4) = 4$. $P(\text{neither}) = 4/20 = 1/5$.

F Foundation — Answer: 1/6

Coffee only = $12 - 7 = 5$. $P(\text{coffee only}) = 5/30 = 1/6$.

F Foundation — Answer: 23/25

Maths only = $18 - 10 = 8$. English only = $15 - 10 = 5$. Both = 10. Total studying at least one = $8 + 5 + 10 = 23$. $P(\text{at least one}) = 23/25$.

C Core — Answer: 8

Let x = both. Biology only = $22 - x$, Chemistry only = $18 - x$. Total = $(22 - x) + x + (18 - x) + 8 = 40$. So $48 - x = 40$, $x = 8$.

C Core — Answer: 7/10

Cats only = $30 - 10 = 20$. Dogs only = $25 - 10 = 15$. Exactly one = $20 + 15 = 35$. $P(\text{exactly one}) = 35/50 = 7/10$.

C Core — Answer: 1/4

Let x = both. A only = $35 - x$, B only = $28 - x$. Total = $(35 - x) + x + (28 - x) + 12 = 60$. So $75 - x = 60$, $x = 15$. $P(\text{both}) = 15/60 = 1/4$.

C Core — Answer: 1/5

$A \cap B = \{4, 5\}$. $n(A \cap B) = 2$. Total = 10. $P = 2/10 = 1/5$.

C Core — Answer: 7/12

$P = \{3, 6, 9, 12\}$, $Q = \{1, 2, 3, 4, 6, 12\}$. $P \cup Q = \{1, 2, 3, 4, 6, 9, 12\}$. $n(P \cup Q) = 7$. Total = 12. $P = 7/12$.

C Core — Answer: 2/5

Total = 40. Neither = 5. So at least one sport = 35. Let x = both. $(25 - x) + x + (20 - x) = 35$, $x = 10$. $P(\text{football} | \text{cricket}) = 10/25 = 2/5$.

E Unknown — Answer: 0

$A \cap B \cap C$ = numbers even, multiples of 3, and 5 = multiples of $\text{LCM}(2, 3, 5) = 30$. No multiples of $30 \leq 20$. So empty set. $P = 0$.

E Unknown — Answer: 11/40

History only = $42 - 28 = 14$. Geography only = $36 - 28 = 8$. Exactly one = $14 + 8 = 22$. $P(\text{exactly one}) = 22/80 = 11/40$.

E Unknown — Answer: 0.2

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$. So $0.8 = 0.6 + 0.4 - P(A \cap B)$. So $P(A \cap B) = 1.0 - 0.8 = 0.2$.

E Unknown — Answer: 0.35

$P(A \cup B) = 0.5 + 0.3 - 0.15 = 0.65$. $P(A' \cap B') = P((A \cup B)') = 1 - 0.65 = 0.35$.

X Extension — Answer: 3/10

$F+G$ only = $20-5 = 15$. $F+S$ only = $15-5 = 10$. $G+S$ only = $10-5 = 5$. Exactly two = $15+10+5 = 30$. $P = 30/100 = 3/10$.

X Extension — Answer: 0.2

$P(M' \cap P) = P(P) - P(M \cap P) = 0.6 - 0.4 = 0.2$.

G1_Q6: Expected Value - Problems

F Foundation — Answer: 10

$P(4) = 1/6$. Expected number = $60 \times 1/6 = 10$.

F Foundation — Answer: 30

$P(\text{red}) = 3/8$. Expected number = $80 \times 3/8 = 30$.

F Foundation — Answer: 25

$P(7) = 1/8$. Expected = $200 \times 1/8 = 25$.

F Foundation — Answer: 150

$P(\text{heads}) = 1/2$. Expected = $300 \times 1/2 = 150$.

C Core — Answer: \$0.10

$E = 5 \times 0.3 + (-2) \times 0.7 = 1.5 - 1.4 = 0.10$. Expected value = 0.10.

C Core — Answer: 60

Numbers > 4 : 5 and 6. $P(>4) = 2/6 = 1/3$. Expected = $180 \times 1/3 = 60$.

C Core — Answer: 65

$P(\text{heart}) = 13/52 = 1/4$. Expected = $260 \times 1/4 = 65$.

C Core — Answer: \$0.95

Total prize money = $50 + 2 \times 10 + 5 \times 5 = 50 + 20 + 25 = 95$. Expected winnings = $95/100 = 0.95$.

C Core — Answer: 60

Total = 10. $P(\text{white}) = 3/10$. Expected = $200 \times 3/10 = 60$.

C Core — Answer: -\$1/3

$P(6)=1/6$, win 6. $P(1)=1/6$, win 4. $P(2,3,4,5)=4/6=2/3$, win 0. Expected winnings = $(1/6) \times 6 + (1/6) \times 4 = 6/6 + 4/6 = 10/6 = 5/3 \approx 1.67$. Cost = 2. Expected profit = $5/3 - 2 = 5/3 - 6/3 = -1/3$.

E Unknown — Answer: \$11

Expected winnings = $(1/5) \times 1^2 + (1/5) \times 2^2 + (1/5) \times 3^2 + (1/5) \times 4^2 + (1/5) \times 5^2 = (1+4+9+16+25)/5 = 55/5 = 11$.

E Unknown — Answer: 12

Total outcomes = 36. Sum = 5: (1,4),(2,3),(3,2),(4,1) — 4 ways. $P(\text{sum}=5) = 4/36 = 1/9$. Expected = $108 \times 1/9 = 12$.

E Unknown — Answer: -\$1.00

Expected winnings = $100 \times 0.02 + 20 \times 0.05 + 10 \times 0.1 = 2 + 1 + 1 = 4$. Cost = 5. Expected profit = $4 - 5 = -1$. So expected loss of 1.

E Unknown — Answer: \$0.00

$P(\text{red}) = 4/10 = 2/5$. $P(\text{blue}) = 6/10 = 3/5$. Expected value = $3 \times (2/5) + (-2) \times (3/5) = 6/5 - 6/5 = 0$.

X Extension — Answer: 25/9

$P(\text{sum}=7) = 6/36 = 1/6$. $P(\text{sum}=2) = 1/36$. $P(\text{sum}=12) = 1/36$. Expected winnings = $10 \times (1/6) + 20 \times (1/36) + 20 \times (1/36) = 10/6 + 40/36 + 40/36 = 100/36 = 25/9$. For a fair game, $c = \text{expected winnings} = 25/9$.

X Extension — Answer: -\$464/221

$C(52,2) = 1326$. $P(\text{both aces}) = C(4,2)/1326 = 6/1326 = 1/221$. $P(\text{exactly one ace}) = (4 \times 48)/1326 = 192/1326 = 32/221$. Expected winnings = $100 \times (1/221) + 10 \times (32/221) = (100 + 320)/221 = 420/221$. Cost = 4. Expected profit = $420/221 - 4 = 420/221 - 884/221 = -464/221$.

G1_Q7: Experimental vs Theoretical Probability - Problems

F Foundation — Answer: 14/25

Experimental probability = number of heads / total tosses = $28/50 = 14/25$.

F Foundation — Answer: 2/15

Experimental probability = $8/60 = 2/15$.

F Foundation — Answer: 3/5

Experimental probability = $120/200 = 3/5$.

F Foundation — Answer: 3/8

Experimental probability = $15/40 = 3/8$.

C Core — Answer: 17/100

Total = 200. Frequency of 6 = 34. Experimental probability = $34/200 = 17/100$.

C Core — Answer: 11

Experimental probability = frequency / total. So frequency = $(11/60) \times 60 = 11$.

C Core — Answer: 180

Experimental probability = $72/200 = 9/25 = 0.36$. Estimated number in 500 tosses = $500 \times 9/25 = 180$.

C Core — Answer: 3/100

Theoretical probability = $1/2 = 50/100$. Experimental probability = $53/100$. Difference = $53/100 - 50/100 = 3/100$.

C Core — Answer: 2/5

Experimental probability = $32/80 = 2/5$.

C Core — Answer: 57/100

Odd numbers: 1,3,5. Total odd frequency = $98+95+92 = 285$. Total spins = 500. Experimental probability = $285/500 = 57/100$.

E Unknown — Answer: 16%

Theoretical = $1/6 \approx 0.1667$. Experimental = $42/300 = 0.14 = 7/50$. Percentage error = $((1/6 - 7/50)/(1/6)) \times 100\% = ((25/150 - 21/150)/(25/150)) \times 100\% = (4/25) \times 100\% = 16\%$.

E Unknown — Answer: 180

Experimental probability of heads = $32/80 = 2/5 = 0.4$. Estimated heads in 450 tosses = $450 \times 2/5 = 180$.

E Unknown — Answer: No, $0.32 \neq 0.2$

If fair, $P(\text{blue}) = 1/5 = 0.2$. The experimental probability is 0.32, which is significantly different from 0.2, suggesting the spinner may be biased.

E Unknown — Answer: 66

Primes on a dice: 2,3,5. Frequency = experimental probability $\times$ total = $0.55 \times 120 = 66$.

X Extension — Answer: As n increases, experimental probability approaches theoretical probability (0.5)

The Law of Large Numbers states that as the number of trials increases, the experimental probability tends to converge to the theoretical probability. Initially with 10 tosses, 7 heads (0.7) is possible due to random variation. With 1000 tosses, the proportion stabilises near 0.5, giving 0.501.



Extension — Answer: 22

Theoretical $P(\text{sum}=7)=6/36=1/6$. Current: $28/180=7/45$. Target: after 300 total rolls, $x/300=1/6$, so $x=50$. Additional sum7 needed = $50-28 = 22$.